Introduction to Data Science and R Programming

UNIT – I: Introduction to Data Science

1. Define Data Science. What are its key components?

2. Explain the lifecycle/phases of a Data Science project.

3. What is the difference between Data Science, Machine Learning, and Artificial Intelligence?

4. Describe the roles and responsibilities of a Data Scientist.

5. Explain the types of data: Structured, Semi-structured, and Unstructured.

UNIT – II: Data Collection and Preprocessing

1. What are the methods of data collection in data science?

2. Explain the importance and techniques of data cleaning.

3. What is data preprocessing? List its steps.

4. Define and explain data normalization and standardization.

5. Explain outlier detection and handling techniques.

UNIT – III: Introduction to R Programming

1. What is R? List its features and applications in Data Science.

2. Explain data types in R with examples.

3. Write R programs to demonstrate use of:

   a) Vectors

   b) Lists

   c) Matrices

4. How to create and manipulate data frames in R?

5. What are factors in R? Explain with example.

UNIT – IV: Control Structures and Functions in R

1. Explain conditional statements in R (if, if...else, switch) with examples.

2. Discuss looping structures in R (for, while, repeat) with examples.

3. How to create user-defined functions in R? Give an example.

4. What is the use of built-in functions like mean(), sum(), length(), max(), min()?

5. Explain the scope of variables in R functions.

UNIT – V: Data Visualization and File Handling in R

1. Explain how to read data from CSV files in R.

2. How to write data to external files (CSV) using R?

3. Explain different types of plots in R (barplot, hist, boxplot, scatter plot).
4. Write R code to generate a pie chart with labels and colors.
5. Discuss the use of ggplot2 library in R with a simple example.